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Verify Before You Substitute: Safe Generic Medicine Recommendation from Medicine Strip Images

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Verify Before You Substitute: Safe Generic Medicine Recommendation from Medicine Strip Images

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***Abstract*—Patients in India often pay very different prices for the same medicine, depending on the brand. Cheaper generic equivalents exist, but the composition printed on a blister strip is hard to read and a wrong match can cause harm. We present GeneriScan, a method that reads a photograph of a medicine strip with two independent optical character recognition (OCR) engines, matches both readings against 166,972 Indian tablet and capsule products, and suggests a cheaper generic only when the salts, strengths and dosage form agree. Otherwise it abstains. A small set of fixed rules flags narrow therapeutic index drugs. In a pilot evaluation on 17 public photographs of 9 Indian products, the method answered for 6/17 images and made no unsafe suggestion, while brand matching alone made one unsafe suggestion in five answers and naive substitution was unsafe in all five. For the five verified products with a known brand price, the suggested equivalents cost 71% less on average. The sample is small, and we set out the larger evaluation needed to confirm these findings.**

***Keywords*—optical character recognition, medicine identification, generic medicines, selective prediction, medication safety, healthcare informatics**

I. Introduction

A large share of health spending in India is paid directly by households, and medicines make up much of it. Families managing diabetes, hypertension or thyroid disorders buy the same tablets every month, often for years. The same molecule is sold under many brand names, and prices for an identical salt and strength can differ several times over. The Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP) sells generics of assured quality through dedicated outlets at lower prices [1], yet many patients do not know which generic matches the brand they were prescribed.

Two practical problems stand in the way. First, the composition is printed in small type on curved, reflective foil, and part of the strip is often torn away once tablets have been taken. Second, equivalence is easy to get wrong. Metformin 500 mg and metformin 500 mg sustained release are not interchangeable, and combination tablets with two or three salts multiply the chances of error. For drugs with a narrow therapeutic index (NTI), even a switch to another brand of the same salt should be approved by a doctor [2]. A tool that guesses is therefore worse than no tool at all.

Academic work has mostly identified known strips or restored expiry dates [3], [4]. Several open-source applications go further and map a strip photograph to cheaper generics, including MedLens [5], Aushadhi [6] and MediScan [7]. They show that the idea is practical. To our knowledge, however, none of them reports how often a suggested medicine differs from the original in salt, strength or form when tested on real photographs. This paper treats the task as safe selective prediction. Our contributions are as follows.

1) A method that combines two independent readings of a strip with a reference database and declines to answer when they disagree.
2) An evaluation protocol that reports the unsafe suggestion rate alongside accuracy and coverage, released as an open script.
3) A pilot study on 17 real photographs of Indian medicine strips, comparing the method with OCR, brand matching and naive substitution baselines.
4) An estimate of the savings available to patients when a verified equivalent exists.

Section II reviews related work and Section III describes the data. Section IV presents the method and Section V the experimental setup. Section VI reports results, Section VII discusses limitations and Section VIII concludes.

II. Related Work

A. Medicine Recognition and Substitution

Identifying medicines from images has mostly been treated as a classification problem. The NLM Pill Image Recognition Challenge matched photographs of loose tablets against reference images [8]. For blister strips, Rout *et al.* used cepstral features computed in two dimensions with k-nearest neighbour, support vector machine and logistic regression classifiers [3]. Such methods recognise products seen during training but cannot read the composition of a new one. Moorthy *et al.* restored blurred expiry dates on strips with a residual network [4], and a study of antibiotic packaging in Thailand linked OCR confidence to reflectance and print quality [9].

Open-source tools now suggest substitutes directly. MedLens reads strips with PP-OCR models, maps the text to about 254,000 Indian medicines with a neural retriever and fuzzy reranking, and lists cheaper brands with identical salts, strengths and form; its published benchmark uses synthetic inputs [5]. Aushadhi asks a large language model to read a strip or prescription and returns the exact Jan Aushadhi generic or no match at all [6]. MediScan combines YOLOv8 detection with TrOCR to extract salts and compute generic alternatives [7]. Our work differs in requiring two independent readings to agree before answering, and in measuring unsafe suggestions on real photographs.

B. Text Extraction from Images

Classical OCR engines such as Tesseract [10] and lightweight deep pipelines such as PP-OCR [11] detect and recognise text but leave its interpretation to later rules. Models that also use page layout, such as LayoutLM [12], and models that skip OCR altogether, such as Donut [13], read structured documents directly. General vision-language models such as Qwen2-VL [14] can answer questions about text in natural images, but they can also produce fluent output that is not present in the image.

C. Selective Prediction

A classifier that may decline to answer trades coverage for accuracy. Chow formalised this reject option [15], and Geifman and El-Yaniv extended it to deep networks through risk–coverage curves [16]. In our setting a missed suggestion costs the patient money, while a wrong one may cause harm, so we report the unsafe suggestion rate at each level of coverage.

D. Medication Safety

The World Health Organization’s Medication Without Harm initiative identifies confusion between products as a recurring cause of avoidable harm [17]. Bioequivalence limits for NTI drugs are tighter than for other drugs [2], which is why we leave brand switching for these drugs to the prescriber.

III. Data

A. Pilot Evaluation Set

The pilot uses photographs already published in open-source repositories. Sixteen images come from the test set of a strip identification project [18]: the front and the back of eight strips sold in India. They appear to be e-pharmacy catalogue photographs, taken flat and evenly lit, with a watermark. One further image is a smartphone photograph of a levocetirizine strip from the MediScan repository [7]. We excluded a handwritten prescription, a box for an invented brand, and 12 synthetic renders whose printed composition contradicted the real brand. The backs show only tablets, so a correct system should abstain on them. The images are used for evaluation only and are not redistributed.

B. Ground Truth

For each image we transcribed the salts, strengths and dosage form printed on the strip and checked them against the labels supplied with the source repository. Two folders were named by salt; the printed brands are Lomofen Plus and Levocetrizen 5. The methods need no training, so all 17 images form the test set, and every threshold and weight was fixed before the run. Table I summarises the set.

TABLE I
PILOT EVALUATION SET

Property	Value
Images	17
Distinct products	9
One salt / combination products	7 / 2
Modified release products	0
Front, catalogue / back, catalogue / smartphone	8 / 8 / 1
Images with the composition visible	9

C. Reference Database

Brand compositions come from the Indian Medicine Dataset [19], an MIT-licensed collection of 253,973 products with prices recorded in November 2022. After removing discontinued products and keeping tablets and capsules with a readable composition, 166,972 rows remain. Generic prices come from the PMBJP product list of 2,439 items [1], as distributed with [6] and retrieved on 10 October 2026; 1,323 of these are tablets or capsules with complete strengths. Salt names were normalised by removing pharmacopoeial suffixes (IP, BP, USP) and the words hydrochloride and dihydrochloride. Other salt forms, such as succinate and tartrate, were kept because they change the product.

IV. Proposed Method

A. Overview

Fig. 1 shows the pipeline. A photograph is read by two independent OCR engines. Both readings are used to retrieve candidate products from the reference database, and a candidate is accepted only if its salts, strengths and dosage form agree with at least one reading. Accepted products pass through the safety rules before cheaper equivalents are listed.

If no candidate passes, the system abstains and asks for another photograph.

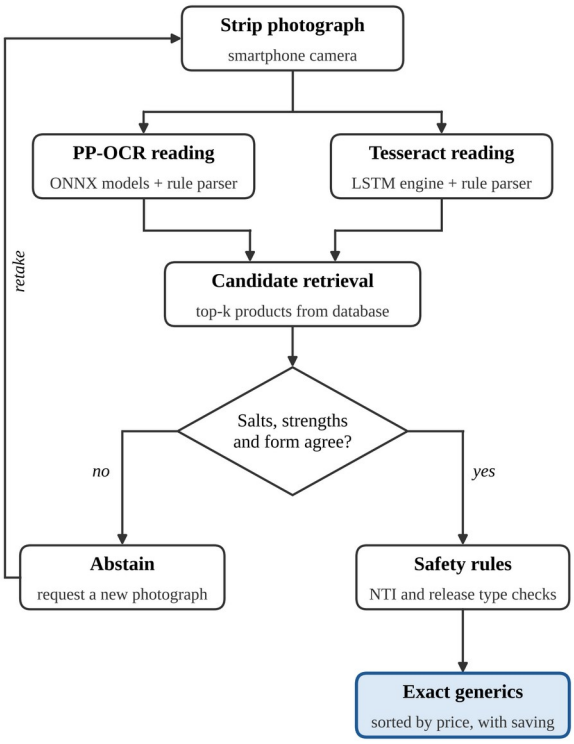


Fig. 1. Overview of the GeneriScan pipeline. A generic is suggested only after a reading has been verified against the reference database; otherwise the user is asked to retake the photograph.

B. Two Independent Readings

The first reader runs the PP-OCR detection and recognition models [11] through RapidOCR on ONNX Runtime [20]. The second is Tesseract 5 with its LSTM engine [10], applied to a greyscale copy enlarged so that its longer side is 1,500 pixels. Both readings pass through the same parser. It corrects drug words within a small edit distance of a known salt, finds salt names from the database vocabulary, attaches to each salt the first strength printed within 45 characters after it, and detects release type codes such as SR, ER, MR and CR. The two engines fail in different ways, which is what makes their agreement informative. A vision-language model could replace either reader; we leave that comparison to future work.

C. Candidate Retrieval

For each reading, every text line is compared with all brand names using normalised Levenshtein similarity [21], and the five closest brands with a similarity of at least 0.85 are kept. Products whose composition exactly matches a reading are added as well. The union over both readings forms the candidate set P.

D. Verification and Abstention

Let a candidate p have salt set $S(p)$, strength vector $v(p)$ and form $f(p)$. A reading r is *consistent* with p, written $\delta(r, p) = 1$, when its salts, strengths and form all match exactly after normalisation, and $\delta(r, p) = 0$ otherwise. We score each candidate as

$$c(p) = \alpha\, s_b(p) + \beta\, \delta(r_1, p) + \gamma\, \delta(r_2, p) \qquad (1)$$

where s_b is the similarity between brand names. The system accepts the highest scoring candidate $p*$ only if at least one reading is consistent with it and $c(p*) \geq \tau$. We set $\alpha = \beta = \gamma = 0.5$ and $\tau = 0.5$ before the experiment, so in practice a candidate is accepted when one reading agrees with it fully.

E. Safety Rules

Suggestions containing an NTI drug, such as levothyroxine, warfarin, phenytoin, carbamazepine, valproate, lithium or digoxin, carry a warning to switch only on a doctor's advice [2]. Substitution is never offered across release types or between combinations with different salts. The system gives no dosing advice, and every result carries a notice that a doctor or pharmacist should confirm any change.

F. Substitution and Savings

For a verified product $p*$, the system lists products with the same salts, strengths and form, preferring a PMBJP generic and otherwise the cheapest market brand by price per tablet. The saving per strip is the difference in price per tablet multiplied by the strip size of $p*$.

V. Experimental Setup

A. Baselines

B1 parses the Tesseract text with the same rules and suggests the cheapest product with the parsed composition. B2 takes the brand line from PP-OCR that best matches the database (similarity of at least 0.85) and trusts that brand's recorded composition. B4 takes the brand found by B2 and suggests the cheapest product sharing one of its salts, without checking strength or form; it shows what happens when no verification is done. A vision-language baseline was planned but could not be run: the experiments used a two-core CPU with 7 GB of memory and no access to model repositories.

B. Metrics

Salt, strength and form accuracy are the share of the nine images with a visible composition on which that field is entirely correct; *exact composition accuracy* requires all three. Let $N = 17$ be the number of test images, A the set of images on which a method gives a suggestion, and U the set of suggestions that differ from the true product in salt, strength or form. Then

$$\text{Coverage} = |A| / N, \quad \text{Unsafe rate} = |U| / |A| \qquad (2)$$

Because the sample is small, we report counts and give Wilson 95% confidence intervals for the main rates.

C. Implementation

All code is in Python 3.11 and uses RapidOCR 1.2.3 with ONNX Runtime 1.29.0, Tesseract 5.3.4 and RapidFuzz 3.14.6. Running both OCR engines took 1.58 s per image on average. The evaluation script, the ground truth and the per-image results are available at [repository link].

VI. Results and Discussion

A. Main Results

Table II gives the results. The proposed method answered for 6/17 images and none of its six suggestions was unsafe (95% CI 0–39%). B2 answered for five images, one of them unsafe (20%, CI 4–62%). Naive substitution turned every one of B2's five answers into an unsafe suggestion, because the cheapest product sharing a salt usually had a different strength. On the nine images with a visible composition, exact extraction was 6/9 for the proposed method, 4/9 for B2 and 3/9 for B1. The differences point the expected way, but the intervals overlap and a larger test set is needed.

Table II
Results on the Pilot Set (Counts)

Method	Salt	Str.	Form	Exact	Cov.	Unsafe
B1: Tesseract	5/9	5/9	3/9	3/9	4/17	1/4
B2: brand match	4/9	4/9	5/9	4/9	5/17	1/5
B4: naive subst.	4/9	4/9	5/9	4/9	5/17	5/5
Proposed	6/9	6/9	6/9	6/9	6/17	0/6

Salt, strength, form and exact accuracy are out of the nine images with a visible composition; coverage is out of all 17 images; unsafe counts are out of the suggestions made.

B. Coverage and Safety

Fig. 2 plots the unsafe suggestion rate against coverage as the acceptance threshold is lowered. Abstention explains most of the difference between the methods. Every method answered nothing on the eight backs. The proposed method also declined three fronts: two where the parser could not attach a strength to the salt, and the low resolution Gelusil strip.

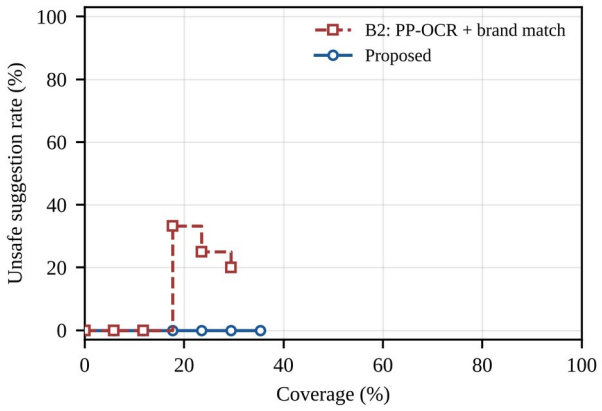


Fig. 2. Unsafe suggestion rate against coverage on the 17 pilot images, as the acceptance threshold is lowered. B2 reaches its first unsafe answer at 18% coverage; the proposed method stays at zero up to its full coverage of 35%.

C. Ablation Study

Table III
Ablation of the Proposed Method (Counts)

Variant	Exact	Coverage	Unsafe
Full method	6/9	6/17	0/6
Without Tesseract	6/9	6/17	0/6
Without PP-OCR	3/9	3/17	0/3
Without strength check	8/9	8/17	0/8
Without form check	7/9	7/17	0/7

Removing Tesseract changed nothing on this set, while removing PP-OCR halved coverage, so the second reader did not yet earn its place here. The strength check cost two correct answers, Dolo 650 and Metrogyl 400, where the parser could not attach a strength to the salt from the OCR text. The form check cost one, Metrogyl 400, where noise in the Tesseract text resembled a release type code. Neither check prevented an unsafe answer in this sample. We keep both because B4 shows how easily a strength mismatch arises without verification, but their benefit must be shown on a larger set.

D. Effect of Capture Conditions

Table IV
Results by Capture Condition

Condition	Images	B2: exact / unsafe	Proposed: exact / unsafe
Front, catalogue photo	8	4 / 1	5 / 0
Back, no printed text	8	0 / 0	0 / 0
Smartphone photo	1	0 / 0	1 / 0

Both methods abstained on all eight backs, which is the correct behaviour. On the fronts the proposed method read one more composition correctly and avoided B2's unsafe answer. It also handled the single smartphone photograph, where B2 found no brand line close enough to the database.

E. Potential Savings

Table V
Price per Tablet and Saving per Strip (₹)

Product	Brand	Equivalent	Strip saving	Saving (%)
Acenext P	5.80	1.03 (PMBJP)	47.68	82
Brufen 400	1.29	0.61 (PMBJP)	10.24	53
Levipil 500	12.87	6.98 (PMBJP)	88.34	46
Lomofen Plus	2.44	0.38 (Market)	41.30	85
Levocetrizen 5	3.50	0.31 (Market)	31.88	91

Table V lists the five products for which a verified equivalent was found and the brand price was known. The mean saving was ₹43.89 per strip (median ₹41.30), or 71% of the brand price. Three equivalents came from the PMBJP list and two from the cheapest market brand, as no matching PMBJP tablet was found in the parsed list. Brand and market prices date from November 2022 and PMBJP prices from October 2026, so these figures are indicative only.

F. Error Analysis

Fig. 3 shows the clearest failure of brand matching. On the Acenext P strip, both OCR readings found aceclofenac 100 mg and paracetamol 325 mg, but B2 matched a product named "Paracetamol Tablet" and suggested paracetamol 500 mg alone. The proposed method accepted the composition that both readings supported. B1 made one unsafe suggestion: on the Levocetrizen 5 strip, noise in the Tesseract text looked like an SR code, so it offered a sustained release product. The proposed method missed Dolo 650 because PP-OCR read the composition line as "Porocetamol iP650mg", too far from "paracetamol" for the spelling correction, and it missed Metrogyl 400 because the salt was read as "MetronigazoleIP". Tesseract read little from the foil of either strip. The Gelusil MPS front was only 480 × 281 pixels, and one of its four salts, magnesium aluminium silicate, is missing from the reference vocabulary.

Field	PP-OCR reading	Tesseract reading	B2 brand match
Salts	Aceclofenac + Paracetamol	Aceclofenac + Paracetamol	Paracetamol
Strengths	100 mg + 325 mg	100 mg + 325 mg	500 mg
Form	tablet	tablet	tablet

B2 (brand match): unsafe
Matched the product "Paracetamol Tablet" and suggested Paracetamol 500 mg, dropping aceclofenac.

Proposed: safe
Both readings agree with the same composition; suggests Jan Aushadhi aceclofenac 100 mg + paracetamol 325 mg.

Fig. 3. A real case from the pilot run (Acenext P front). Brand matching trusts a wrongly matched product; the proposed method accepts only the composition that the readings support.

VII. Limitations and Ethical Considerations

These results come from a pilot of 17 images, 16 of them catalogue photographs rather than photographs taken by patients, and the confidence intervals are wide. No product in the set is modified release or NTI, so the form check and the NTI warning were not tested on positive cases. The vision-language reader described as an alternative was not evaluated. Ground truth was transcribed by the authors from the printed text. Equivalence is judged from composition alone; we did not assess bioequivalence, which cannot be inferred from packaging. Prices change over time, and the brand prices used are from 2022.

GeneriScan is meant to support decisions, not to make them. It does not alter prescriptions, gives no dosing advice, and asks the user to confirm any change with a doctor or pharmacist. Abstention is a deliberate design choice: when the system is unsure, it says so rather than guessing. No patient data were used, and the evaluation images are not redistributed.

VIII. Conclusion and Future Work

We presented GeneriScan, a method that reads a medicine strip with two independent OCR engines and suggests a cheaper generic only after verifying the salts, strengths and form against a reference database. On a pilot of 17 real photographs it made no unsafe suggestion in six answers, while brand matching made one in five and naive substitution was unsafe in all five. Verified equivalents cost 71% less on average.

The next step is a larger test set of 300 to 500 smartphone photographs taken under everyday conditions, with modified release and NTI products, which our group is now collecting. We will also test a vision-language model as one of the two readers, and add an interface in Hindi and Kannada with spoken output for elderly users and those with limited literacy.

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